



**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY
LONERE - RAIGAD - 402 103**

Summer Semester Examination - May - 2019

Course: B. Pharm.

Semester: IV

Subject with Subject Code: Pharmaceutical Organic Chemistry-III (BP401T)

Date: 27/05/2019

Marks: 75

Duration: 3hrs

Instructions: i) All questions are compulsory
ii) Figures to the right indicate full marks
iii) Draw the diagrams or flow charts wherever necessary.

Q. 1 Choose the correct alternative.

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1. A molecule is said to be chiral if-
 - A. It has four different atoms attached to carbon
 - B. Non super imposable on its mirror image
 - C. Superimposable on its mirror image
 - D. Both A & B
2. Specific rotation is denoted by-
 - A. R
 - B. S
 - C. D
 - D. $[\alpha]$
3. Angle strain is maximum in-
 - A. Cyclopropane
 - B. Cyclobutane
 - C. Cyclopentane
 - D. Cyclohexane
4. Racemic mixture can be separated by-
 - A. Mechanical Separation
 - B. Biochemical separation
 - C. Chemical Method
 - D. All of the above
5. Stereoisomers which are not mirror images of each other are called-
 - A. Enantiomers
 - B. Diastereomers
 - C. Racemic mixtures
 - D. None of the above
6. Pyridine is basic in nature-
 - A. Due to presence of lone pair of electrons on nitrogen atom
 - B. Due to formation of salts with acid
 - C. Due to formation of quaternary salt with alkyl halide
 - D. All of the above
7. Electrophilic substitution reactions occurs in Quinoline at-
 - A. Position 2 & 4
 - B. Position 3 & 5
 - C. Position 5 & 8
 - D. Position 4 & 6
8. Which of the ring in isoquinoline gets easily reduced?
 - A. Benzene ring
 - B. Nitrogen containing ring
 - C. Both rings
 - D. None of the above
9. In indole which of the heterocyclic ring is fused with benzene ring-
 - A. Pyrroazole
 - B. Imidazole
 - C. Isoxazole
 - D. Pyrrole

10. Sulphonation of indole in presence of pyridine occurs at-

- A. Position 2
- B. Position 1
- C. Position 3
- D. Position 4

11. In acridine which of the following catalyst causes reduction of benzene ring?

- A. Zn/HCl
- B. Pt/HCl
- C. LiAlH_4
- D. Both A & C

12. In thiazole and oxazole nucleophile and electrophile attacks at-

- A. C1 & C2 respectively
- B. C2 & C3 respectively
- C. C2 & C3 respectively
- D. C1 & C5 respectively

13. In pyrrole the number of electrons that the N atom contributes to the π -system is-

- A. 2
- B. 1
- C. 3
- D. 4

14. Electrophilic substitution in furan usually occurs at-

- A. C3 atom
- B. C2 atom
- C. Both C2 & C3 atoms
- D. None of the above

15. Which statement about thiophene is correct?

- A. Thiophene is polar
- B. Oxidative polymerization of thiophene leads to a conjugating polymer
- C. Thiophene is more reactive towards electrophiles than furan
- D. The S atom contributes two electrons to the π -system.

16. Which of the following is strong hydride donor?

- A. NaBH_4
- B. LiAlH_4
- C. B_2H_6
- D. None of the above.

17. Oppenauer oxidation is the reverse process of-

- A. Wolff-Kishner reduction
- B. Clemmensen reduction
- C. Meerwein-Ponndorf-Verley reduction
- D. Rosenmund's reduction

18. Birch reduction is done by using-

- A. Sodium in ammonia with alcohol
- B. Potassium in ammonia with alcohol
- C. Magnesium in ammonia with alcohol
- D. None of the above.

19. Beckmann's rearrangement is useful for synthesis of-

- A. Alcohols
- B. Amide
- C. Phenols
- D. Oxime

20. In Dakin reaction the $-\text{OH}$ or $-\text{NH}_2$ group must be present at either which position of the reactant-

- A. Ortho position
- B. Meta position
- C. Para position
- D. Both A & C

Q. 2 Answer **any two** of the following questions.

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- A. Give the methods of preparation, chemical reactions and medicinally important compounds for Pyrrole.
- B. Discuss Racemic Mixtures and their Resolution methods.
- C. Explain the reaction & mechanism of Wolf Kishner Reduction.

Q. 3 Answer **any seven** of the following questions.

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- A. Give the Methods of preparation & chemical reactions for the Acridine.
- B. Discuss Methods of determination of configuration of Geometrical isomers.
- C. Give the Structure & medicinally important compounds for (Any-2)
 - a. Thiophene
 - b. Imidazole
 - c. Oxazole
 - d. Indole
- D. Write short note on Oppenaur Oxidation Reaction.
- E. Discuss the confirmations of Butane.
- F. What is Claisen-Schmidt condensation? Give its mechanism & importance.
- G. Define Heterocyclic compounds & classify it.
- H. Explain relative aromaticity & reactivity of pyrrole & Thiophene.
- I. Explain the metal hydride reduction by using Sodium borohydride (NaBH_4).

End